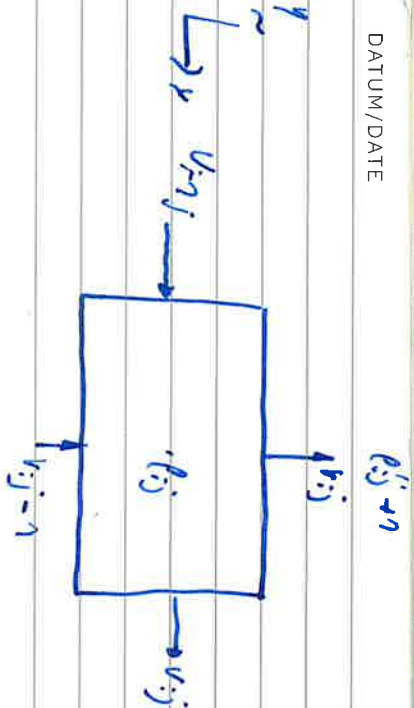
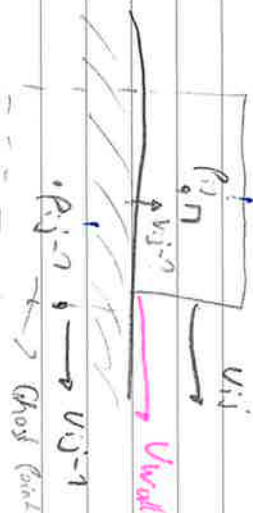




Boundary Conditions

1) No slip at wall Even if mesh is moving



$$\frac{\rho_{i+1,j} + \rho_{i-1,j}}{\Delta x^2} + \frac{\rho_{i,j+1} + \rho_{i,j-1}}{\Delta y^2} - \left(\frac{2}{\Delta x^2} + \frac{2}{\Delta y^2} \right) \rho_{i,j}$$

for wall for wall

$$= \frac{\rho}{\Delta t} \left[\frac{u_{i,j}^* - u_{i,j-1}}{\Delta x} + \frac{u_{i,j} - u_{i,j-1}}{\Delta y} \right]$$

Thus error from projection gives eq. for pressure correction

$$p_{i,j}^*$$

new fluid

$$u_{i,j} = u_{i,j}^* - \frac{\Delta t}{S} \frac{\partial p_{i,j}^*}{\partial x} - p_{i,j}^*$$

now we full

$$u_{i,j}^* = u_{i,j} - \frac{\Delta t}{S} \frac{\partial p_{i,j+1} - p_{i,j}}{\Delta y}$$

continuity should get

$$u_w = \frac{u_{i,j} + u_{i,j-1}}{2}$$

$$u_{i,j-1} = -u_{i,j} + 2u_w$$

$$u_{i,j}^* \text{ projection} = 0$$

→ Von Neumann condition

$$p_{i,j+1} - p_{i,j} = 0 \quad \text{gradient must be 0}$$

$$\frac{\partial p}{\partial n} = 0 \quad \text{at wall}$$

$$\frac{\partial p}{\partial y} = 0 \rightarrow p_{i,j-1} = p_{i,j}$$

Step this condition while programming never cancel bc

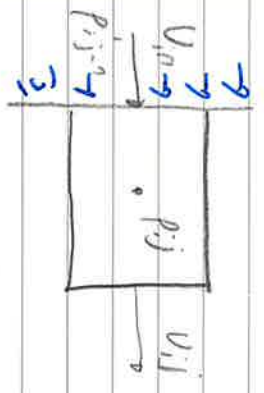
$$\rho_{i,j} \frac{\rho_{i+1,j} - \rho_{i,j}}{\Delta x^2} + \frac{\rho_{i,j+1} + \rho_{i,j-1}}{\Delta y^2} - \left(\frac{2}{\Delta x^2} + \frac{2}{\Delta y^2} \right) \rho_{i,j}$$



$$p_{i+1,j}^{n+1} + \frac{p_{i,j}^{n+1}}{\Delta y} - \left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} \right) p_{ij}^{n+1}$$

$$= \frac{\partial}{\partial t} \left(\frac{v_{ij} - v_{ij}^*}{\Delta x} + \frac{v_{ij}^* - v_{ij}^{n+1}}{\Delta y} \right)$$

2) Inflow condition

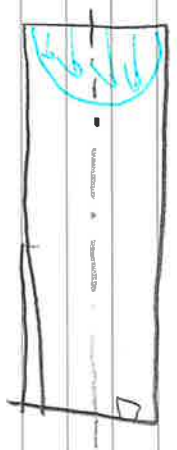


You must not correct the inflow
 $v_{in} = v^* \rightarrow$ Don't and compute the projection / or not you correct it.

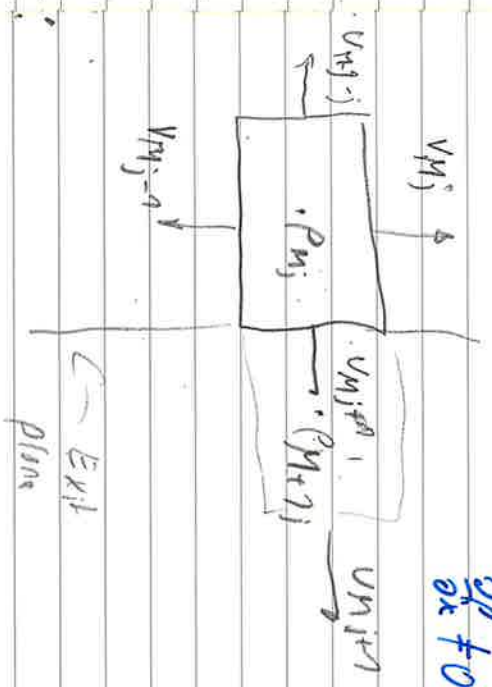
$p_{i,n} = p_{i,j}^{n+1}$ Von Neuman $\rightarrow$ for walls, inlets, outlets

$p_{n+1} = p + p^* \rightarrow$ But Von Neuman allows pressure gradient
 The condition itself is

3) Outflows / Exit BC



$$\frac{\partial v}{\partial x} = 0 \quad \frac{\partial v}{\partial x} = 0$$



$$v_{n+1,j} = v_{n,j} \quad v_{n+1,i} = v_{n,i}$$

$$p_{n+1,j} = p_0 \quad p_{n+1,i} = p_0$$

$p_{n+1,i+j} = 0$... we do not prescribe a gradient
 we cannot compute any correction but need a correction for full continuity

Check for flow in = flow out
 $\partial_i p_{i+1,j} = 0$ set the zero condition by default

if singularity

take out one eq, and set ρ for a point and then solve for ρ , and then the eq.

continuity is always fulfilled, momentum is not.

After it is converged, momentum is also fulfilled.

Given Friedrichs Levy CFL for Runge-Kutta

$$\left[\frac{1}{\Delta x} + \frac{1}{\Delta y} + 2 \left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} \right) \right] \cdot \Delta t$$

$$\Delta t < C^k \cdot 1 \quad C^k \approx 0.5 \text{ for RK4}$$

$\frac{1}{\Delta x} \cdot \Delta t$... perturbation can propagate, may only propagate with the next and not in the (next+1) cell.

CFL for all points and take the minimum allowed timestep.

2026 Seminar Subcommitment

11/18/25

26.7 Michael Subcommitment

X.4 Roger / Peter Subcommitment

X.6 Richard / Ben Subcommitment

6-9.9 So - Michael Conference

Business Meeting for Young Prof.

3.3 9-11:30

10.3 9-11:30

14.3 11.3 9-11:30

24.4 FLIF Consulting Workshop

Freyberg

Cheloni

Roma

Shi: 10.1

28.2 HNNM

24 ZSA-2SA

11.4 GE 90

16.7 Gisseloff / Gisseloff

18.3 TSJ Nonpayment of Research Expenses

14.3 ZG.3

Nach Torsions, von gelb
 Ich bin wieder reingekommen.
 Also ist noch nichts ausgezogen, hat sich aber schon
 weiter gebildet. Ld.

Arbeits ist sehr gut. Ich muss.
 Ich schäufel nicht mehr.
 Flechtling sind ich attraktiv. Dem Torsion hat sich ein
 nur noch präsent. Aber I von kurz aktiv. Torsion
 ist auch so gut mit. Genauer ist Torsion (Eckstein).
 Mariae Verbindung nicht, sie hat viel zu viel Energie, in große
 Schritte.
 Elisabeth hat ein Torsion mit mir & mega bestanden. Dem Torsion.
 Haben gibt ihm Mr. Ld.

Arbeit: Bad weniger spannend nicht schon genommen.
 Anspiel auf 24? Kinder? Kein saureren Erfahrungen
 gemacht. Torsion ungenutzte Pöpel nicht mit an.
 Hat weniger mit 24 geübt, konnte sie da schon von
 schreiben?

Hat heute mir Chris gebracht, von Ludwig
 Was ist das eigentlich für ein Solat?
 " Ups? Solat Solat

Ich habe heute den Kontakt abgesprochen!!!
 Nach Torsion von Torsion erwählt. Will auch 14
 Morgen 5!!!
 Nach 2000 m... mir am 20. hoch sein

Continuity: $\frac{1}{r} \frac{d}{dr} (gr) + \frac{1}{r} \frac{\partial v}{\partial \varphi} = 0$

$\frac{\partial v}{\partial t} + \frac{1}{r} \frac{\partial (vr^2)}{\partial r} + \frac{1}{r} \frac{\partial vr}{\partial \varphi} - \frac{v^2}{r} =$

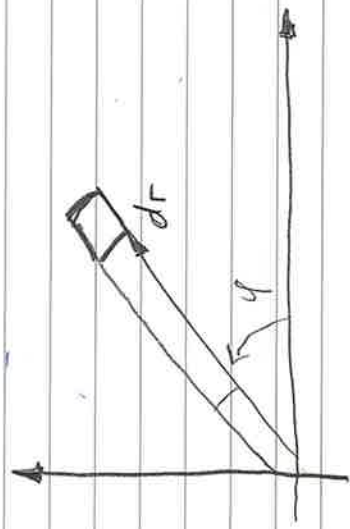
$-\frac{1}{s} \frac{\partial p}{\partial r} + \frac{1}{s} \frac{1}{r} \frac{\partial r}{\partial r} + \frac{1}{s} \frac{\partial \tau_{\varphi\varphi}}{\partial \varphi} - \frac{\tau_{\varphi\varphi}}{r s} + f_r$

$\frac{\partial v}{\partial t} + \frac{1}{r} \frac{\partial (r v r)}{\partial r} + \frac{1}{r} \frac{\partial v^2}{\partial \varphi} + \frac{v v}{r} =$

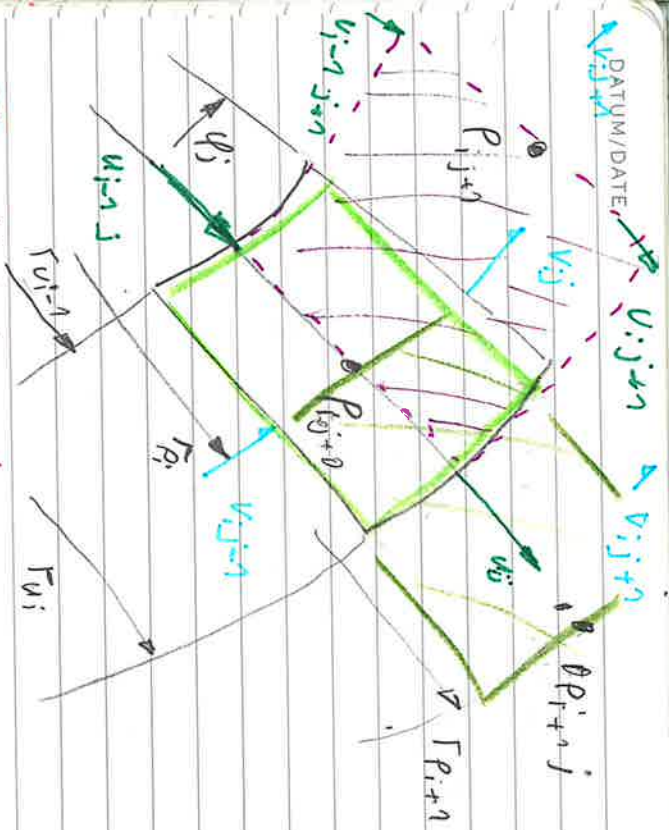
$-\frac{1}{s} \frac{\partial p}{\partial \varphi} + \frac{1}{s} \frac{\partial (r \tau_{r\varphi})}{\partial r} + \frac{1}{s} \frac{\partial \tau_{\varphi\varphi}}{\partial \varphi} + \frac{\tau_{r\varphi}}{s} + \frac{\tau_{\varphi\varphi}}{r} + f_\varphi$

$\tau_{rr} = 2\mu \frac{\partial u}{\partial r}; \tau_{\varphi\varphi} = 2\mu \left(\frac{1}{r} \frac{\partial v}{\partial \varphi} + \frac{u}{r} \right)$

$\tau_{r\varphi} = \mu \left(r \frac{\partial}{\partial r} \left(\frac{v}{r} \right) + \frac{1}{r} \frac{\partial u}{\partial \varphi} \right)$



$dV = r dr d\varphi$



Cell considered for v_{ij}

Shape: For checkerboard problem, pressure can't decay from velocity field

v_{ij} at the face
The pressure is defined at the face, not inside the cell

Open form

Color coding: v_i A Tschur Hydrophobic

Integration of II

→ the cell velocity, cell V

$$\int_{\text{cell}} \frac{\partial v}{\partial t} r dr d\phi \approx \frac{v_{ij}^* - v_{ij}^n}{\Delta t}$$

v_{ij}^* projected v
 $v_{ij}^* \approx v$ at step n

$$P_i (r_{vi} - r_{vi-1}) \cdot (q_{i+1} - q_{i-1}) \cdot \frac{1}{2}$$

$$\int_{\text{cell}} r \frac{\partial (r v)}{\partial r} dr d\phi = \int_{r_{vi-1}}^{r_{vi}} \frac{r v}{r} dr d\phi$$

$$\approx \left[r_{vi} \cdot \left(\frac{v_{i,j+1} + v_{ij}}{2} \right) - \left(\frac{v_{ij} + v_{i,j-1}}{2} \right) \right] \cdot \Delta\phi$$

$$- r_{vi-1} \left(\frac{v_{i,j-1} + v_{i,j-2}}{2} \cdot \frac{v_{ij} + v_{i,j-1}}{2} \right) \cdot \Delta\phi$$

$$\cdot \frac{q_{i+1} + q_{i-1}}{2} \cdot \Delta\phi$$

$$\int_{\text{cell}} r \frac{\partial^2 v}{\partial r^2} r dr d\phi = \int_{r_{vi-1}}^{r_{vi}} \frac{1}{r} \cdot \frac{1}{2} \cdot \Delta\phi$$

$$\approx \left[\left(\frac{v_{ij} + v_{i,j+1}}{2} \right)^2 - \left(\frac{v_{ij} + v_{i,j-1}}{2} \right)^2 \right] \cdot \left[r_{vi} - r_{vi-1} \right] \cdot \Delta\phi$$

Normal flux
momentum

Velocity map

$$\int_{\text{area}} \frac{uv}{r} r dr dq \approx \left[\frac{(u_{i,j+1} + u_{i,j} + u_{i-1,j} + u_{i-1,j+1})}{4} \right]$$

$$\cdot v_{ij} \cdot (q_{i+1} - q_{j-1}) \frac{1}{2} \left[\frac{1}{2} \right]$$

RHS:

$$\int_{\text{area}} \frac{1}{r} \frac{\partial p}{\partial r} r dr dq = \int_{\text{area}} p \left| \frac{q_{i+1} + \frac{1}{2}}{q_j - \frac{1}{2}} \right| dr$$

$$\approx \left(\frac{p_{i,j+1} - p_{i,j}}{1} \right) (r_{ui} - r_{ui-1})$$

Viscous stress

Shear stress

Force balance

$$\frac{1}{S} \int_{\text{area}} \frac{1}{r} \frac{\partial (r \tau_{\theta r})}{\partial r} r dr dq = r_{ui}$$

$$\frac{1}{S} \int_{\text{area}} r \frac{\tau_{\theta r}}{1} dq =$$

$$= \frac{1}{S} \mu \left[\sum \frac{r_{ui}^2}{1} \right]$$

$$\frac{r_{ui-1} v_{i,j+1}}{r_{ui-1} - r_{ui}} - \frac{v_{ij}}{r_{ui}} = \frac{r_{i,j+1} - r_{i,j}}{r_{i,j+1} - r_{i,j}}$$

$$- \left(\frac{r_{i,j+1}}{r_{i,j+1} - r_{i,j}} \right) \frac{v_{ij} - v_{i,j+1}}{r_{i,j+1} - r_{i,j}}$$

$$\left(\frac{v}{r} \right) \frac{1}{dr} \approx v_{i,j+1} \cdot \frac{1}{r_{i,j+1}} \left\{ + \right\}$$

Lower

Upper

$$2 \left(r \cdot \frac{1}{r} \right) \left\{ 2 \frac{v_{i,j+1} - v_{i,j}}{s_{i+1} - q_{j-1}} - 2 \frac{v_{i,j+1} - v_{i,j}}{q_{j+1} - q_{j-1}} \right\}$$

$$\left[(q_{j+1} - q_{j-1}) \cdot \frac{1}{2} \right]$$

Normal stress

$$\int_{\text{area}} \frac{1}{S} \frac{\partial \tau_{\theta r}}{\partial r} r dr dq =$$

$$\frac{1}{S} \int_{\text{area}} \tau_{\theta r} \left| \frac{q_{j+1} + \frac{1}{2}}{q_j - \frac{1}{2}} \right| dr = \frac{1}{S} \sum p_i \left[\frac{v_{i,j+1} - v_{i,j}}{s_{i+1} - q_{j-1}} - \frac{v_{i,j} - v_{i,j-1}}{q_j - q_{j-1}} \right]$$

$$+ \frac{u_{i,j-1} + u_{i,j}}{2} \frac{v_{i,j+1} - v_{i,j}}{2} - \frac{u_{i,j} + u_{i,j+1}}{2} \cdot (r_{ui} - r_{ui-1})$$

Like some for

$$\int_{\text{area}} \frac{1}{S} \frac{\tau_{\theta r}}{r} r dr dq = \int_{\text{area}} \mu \left[\frac{v_{i,j+1} - v_{i,j}}{r_{i,j+1} - r_{i,j}} - \frac{v_{i,j} - v_{i,j-1}}{r_{i,j} - r_{i,j-1}} \right] \cdot \left(\frac{v_{i,j+1} + v_{i,j}}{2 r_{ui}} \right)$$

$$-\left(\frac{2}{r_{i+1} - r_{i-1}}\right) \left(\frac{r_{i+1} - r_{i-1}}{2} \right) (r_{i+1} - r_{i-1})$$

Poisson eq for polar coordinates

Divergence of gradient of pressure gradient

$$\nabla \left(\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial p}{\partial r} \right) \right) = -\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial p}{\partial r} \right)$$

at new + level, uniform continuity: $\nabla \cdot \mathbf{v} = 0$

introduce p into pressure correction

$$\nabla \cdot \mathbf{v} = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial p}{\partial r} \right)$$

new with is calculated coordinates

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial p}{\partial r} \right) + \frac{1}{r} \frac{\partial}{\partial \theta} \left(\frac{1}{r} \frac{\partial}{\partial \theta} \left(r \frac{\partial p}{\partial \theta} \right) \right)$$

$$\frac{1}{r \Delta \theta} \left(\frac{\partial p}{\partial \theta} \right) \Big|_{\theta} \int \int r dr d\theta$$

Orthogonal Poisson eq by terms

$$\iint \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial p}{\partial r} \right) r dr d\theta + \iint \frac{1}{r} \frac{\partial}{\partial \theta} \left(\frac{1}{r} \frac{\partial}{\partial \theta} \left(r \frac{\partial p}{\partial \theta} \right) \right) r dr d\theta =$$

$$\frac{1}{r} \left[\int \int \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial p}{\partial r} \right) r dr d\theta + \int \int \frac{1}{r} \frac{\partial}{\partial \theta} \left(\frac{1}{r} \frac{\partial}{\partial \theta} \left(r \frac{\partial p}{\partial \theta} \right) \right) r dr d\theta \right]$$

$$\int r u \Big|_{r_{i-1}}^{r_i} d\theta + \int \frac{1}{r} v \Big|_{\theta_{j-1}}^{\theta_j} dr =$$

$$\frac{1}{r} \left[\int \frac{\partial p}{\partial r} \Big|_{r_{i-1}}^{r_i} d\theta + \int \frac{\partial p}{\partial \theta} \Big|_{\theta_{j-1}}^{\theta_j} dr \right]$$

$$\left(\frac{r_{i+1} u_{i,j} - r_{i-1} u_{i-1,j}}{r} \right) (r_{i+1} - r_{i-1}) + (v_{i,j} - v_{i,j-1})$$

$$(r_{i+1} - r_{i-1}) = RMS$$

$$\frac{1}{r} \left[\left(\frac{r_{i+1} u_{i,j} - r_{i-1} u_{i-1,j}}{r} \right) (r_{i+1} - r_{i-1}) + (v_{i,j} - v_{i,j-1}) \right]$$

2nd grid points, 2 neighbors cells

Derive V of pressure correction, and

Projection for ρ and ρ' solution for ρ'

or for pressure correction, is obtained as:

$$\rho_{i,j}^{n+1} = \rho - \frac{\Delta t}{S} \frac{\partial p}{\partial r} \quad \text{is defined in shifted V}$$

$$\int_V \frac{\partial p}{\partial r} dV$$

$$\rho_{i,j}^{n+1} = \rho_{i,j}^{n+1} - \rho_{i,j}^{n+1} =$$

$$\left[\rho_{i,j}^{n+1} - \rho_{i,j}^{n+1} \right] (p_i - p_{i-1}) =$$

$$-\frac{\Delta t}{S} \int_V \frac{\partial p}{\partial r} r dr d\phi$$

$$\rho_{i,j}^{n+1} = \rho_{i,j}^{n+1} - \rho_{i,j}^{n+1} = \rho_{i,j}^{n+1} - \rho_{i,j}^{n+1}$$

$$\left[\rho_{i,j}^{n+1} - \rho_{i,j}^{n+1} \right] - \rho_{i,j}^{n+1} = \rho_{i,j}^{n+1} - \rho_{i,j}^{n+1}$$

$$\frac{\Delta t}{S} \rightarrow \rho_{i,j}^{n+1} = \rho_{i,j}^{n+1} - \frac{\Delta t}{S} \frac{\partial p}{\partial r}$$

$$\rho_{i,j}^{n+1} = \rho_{i,j}^{n+1} - \frac{\Delta t}{S} \frac{\partial p}{\partial r}$$

Pressure correction for ρ is obtained in pressure cell.

$$\int_V \frac{\partial p}{\partial r} dV$$

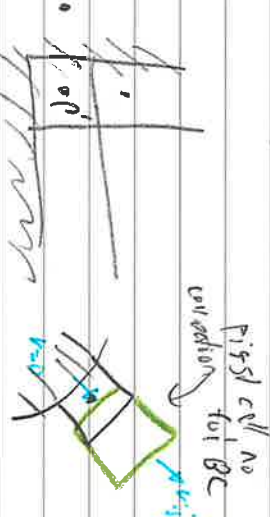
$$\rho_{i,j}^{n+1} = \rho_{i,j}^{n+1} - \frac{\Delta t}{S} \frac{\partial p}{\partial r} = \rho_{i,j}^{n+1} - \rho_{i,j}^{n+1}$$

4th correction: check if continuity is fulfilled & we fulfill it, otherwise only when it is corrected.

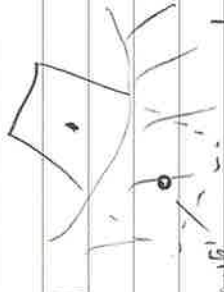
$$\rho_{i,j}^{n+1} = \rho_{i,j}^{n+1} - \rho_{i,j}^{n+1} = \rho_{i,j}^{n+1} - \rho_{i,j}^{n+1}$$

Boundary von Neumann

Fixed wall



We need ghost cells



and $\rho_{i,j}^{n+1} = 0$ stay 0

pressure gradient is 0 then

At flow pressure $\rho' = 0$



Fig. 10.10.1. That out just outside.

Numerical solution of incompressible with projection method (Segregated)

$$\nabla \cdot \mathbf{u} = 0$$

$$\frac{\partial u_i}{\partial t} + \frac{\partial}{\partial x_j} (u_j u_i) = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \tau_{ji} + f_i$$

advection diffusion term

for Newton's

$$\tau_{ij} = 2\mu S_{ij} \quad \text{for incompressible case}$$

$$= \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

Segregated procedure

known level:

$$u_i^n, p^n$$

known level $n+1$

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = -\frac{1}{\rho} \frac{\partial p^n}{\partial x_i} - \{adv\}^n + \{diff\}^n + f_i$$

w/o p → called adv

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = -\frac{1}{\rho} \frac{\partial p^{n+1}}{\partial x_i}$$

$$p^n = p^{n+1}$$

Poisson eq. yields p

1) projection step:

$$u_i^* = u_i^n + \Delta t F \left(\{RHS\} \right)$$

$$1^{st}: F = \{RHS\}$$

only possible with Euler RHS & combine

$$Adams\ Bashfort\ 2^{nd} \quad F = \frac{3}{2} \{RHS\}^n - \frac{1}{2} \{RHS\}^{n-1}$$

2) enforce divergence free new velocity field

$$\frac{\partial u_i^*}{\partial x_i} = 0 \quad \frac{1}{\Delta t} \frac{\partial (u_i^* - u_i^n)}{\partial x_i} = -\frac{\partial}{\partial x_i} \left(\frac{\partial p^{n+1}}{\partial x_i} \right) \frac{1}{\rho}$$

$$\rightarrow \text{Poisson eq. for } p: \quad \frac{\partial^2 p}{\partial x_i \partial x_i} = \frac{\rho}{\Delta t} \frac{\partial u_i^*}{\partial x_i}$$

"Poisson compute step"

$$u_i^{n+1} = u_i^* - \frac{\Delta t}{\rho} \frac{\partial p^{n+1}}{\partial x_i}$$

fulfills mass conservation

steady state $u_i^{n+1} = u_i^n \rightarrow$ fulfills momentum

Discretize in time V of RHS

Cartesian: 2D

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$



$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = \frac{1}{\rho} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + f_x$$

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = \frac{1}{\rho} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + f_x$$

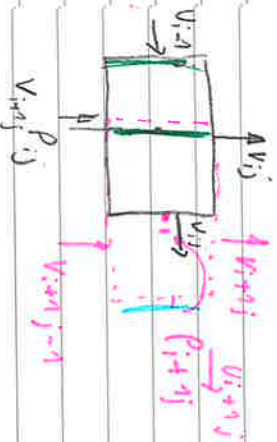
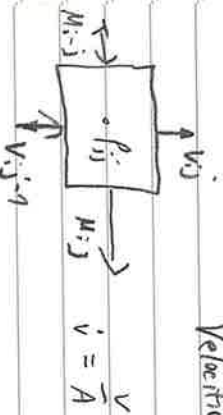
Finite Volume in staggered grid, 2nd order accurate



Control volume

Scalars at center $P_{i,j}$

Velocities are on the faces $\rightarrow$ even to get fluxes



Advancing stencil for u and v

DATUM/DATE

$$\frac{\partial}{\partial t} \iint_M dx dy = \frac{\partial}{\partial t} \bar{M} \Delta x \Delta y$$

Sto. don't compute local values $\rightarrow$ only geometric mean values

$$= \left(\frac{\bar{M}^n - \bar{M}^{n-1}}{\Delta t} \right) \Delta x \Delta y$$

we get from known

$$\iint \frac{\partial u}{\partial x} dx dy = \int_{i-1/2}^{i+1/2} [u] dy$$

Now to approximate the difference of u at 2nd order accurate

$$\approx \left[\left(\frac{u_{i+1,j} + u_{i,j}}{2} \right) - \left(\frac{u_{i,j} + u_{i-1,j}}{2} \right) \right]$$

$$\iint \frac{\partial u}{\partial y} dx dy = \int_{j-1/2}^{j+1/2} [u] dx$$

$$\approx \left[\left(\frac{u_{i,j+1} + u_{i,j}}{2} \right) \cdot \left(\frac{v_{i,j} + v_{i,j+1}}{2} \right) - \left(\frac{u_{i,j} + u_{i,j-1}}{2} \right) \cdot \left(\frac{v_{i,j} + v_{i,j-1}}{2} \right) \right]$$

$$\iint \frac{\partial \rho}{\partial x} dx dy = \int_{i-1/2}^{i+1/2} \rho dy$$

$$\approx \frac{1}{\Delta y} (\rho_{i+1,j} - \rho_{i,j}) \Delta y$$

pressure level in

$$\iint_{\Delta x \Delta y} \frac{\partial^2 u}{\partial x^2} dx dy = \int_{\Delta y} \left[\frac{\partial u}{\partial x} \right]_{i-\frac{1}{2}}^{i+\frac{1}{2}} dy$$

$$\approx \Delta y \left(\frac{u_{i+1,j} - u_{i,j}}{\Delta x} - \frac{u_{i,j} - u_{i-1,j}}{\Delta x} \right) \Delta y$$

$$= \Delta y \left(\frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta x} \right) \Delta y$$

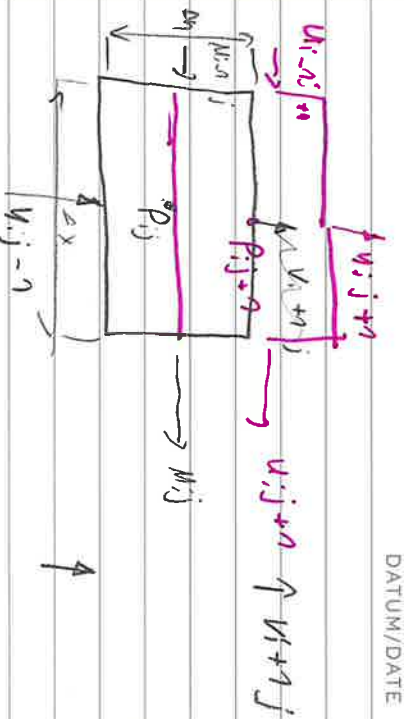
$$\iint_{\Delta x \Delta y} \frac{\partial^2 u}{\partial y^2} dx dy = \int_{\Delta x} \left[\frac{\partial u}{\partial y} \right]_{i,j-\frac{1}{2}}^{i,j+\frac{1}{2}} dx$$

$$\approx \Delta x \left(\frac{u_{i,j+1} - u_{i,j}}{\Delta y} - \frac{u_{i,j} - u_{i,j-1}}{\Delta y} \right) \Delta x$$

$$= \Delta x \left(\frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{\Delta y} \right) \Delta x$$

$$\iint_{\Delta x \Delta y} f(x,y) dx dy \approx \bar{f} \Delta x \Delta y$$

RHS = $\sum$ terms above



$$\iint_{\Delta x \Delta y} \frac{\partial v}{\partial t} = \frac{\bar{v}^* - \bar{v}}{\Delta t} \Delta x \Delta y$$

+ transient term

$$\iint_{\Delta x \Delta y} \frac{\partial}{\partial x} (uv) dx dy = \int_{\Delta y} [uv]_{i-\frac{1}{2}}^{i+\frac{1}{2}} dy$$

$$\approx \left[\frac{u_{i+1,j+1} + u_{i,j}}{2} \cdot \frac{v_{i,j+1} + v_{i,j}}{2} - \frac{u_{i,j+1} + u_{i-1,j}}{2} \cdot \frac{v_{i,j+1} + v_{i,j}}{2} \right] \Delta y$$

objective term

$$\iint_{\Delta x \Delta y} \frac{\partial uv}{\partial y} dx dy = \int_{\Delta x} [uv]_{i,j-\frac{1}{2}}^{i,j+\frac{1}{2}} dx \approx$$

$$\approx \left[\frac{u_{i,j+1} + v_{i,j+1}}{2} - \frac{u_{i,j} + v_{i,j}}{2} \right] \Delta x$$

$$\iint_{\Delta x \Delta y} \frac{\partial p}{\partial y} dx dy = \int_{\Delta x} p \Big|_{i,j-\frac{1}{2}}^{i,j+\frac{1}{2}} dx \approx \frac{p_{i,j+1} - p_{i,j}}{\Delta y} \Delta x$$

$$\iint_{\Delta x \Delta y} \frac{\partial^2 v}{\partial x^2} dx dy = \int_{\Delta y} \frac{\partial v}{\partial x} \bigg|_{dy} =$$

$$\left[\frac{v_{i+1,j} - v_{i,j}}{2} \right]_{\Delta y} = \frac{v_{i+1,j} - 2v_{i,j} + v_{i-1,j}}{\Delta x}$$

$$\iint_{\Delta x \Delta y} \frac{\partial^2 v}{\partial y^2} dx dy = \int_{\Delta x} \left[\frac{\partial v}{\partial y} \right] dx =$$

$$\frac{v_{i,j+1} - 2v_{i,j} + v_{i,j-1}}{\Delta y}$$

$$\frac{\partial^2 p}{\partial x \partial y} = \frac{\partial}{\partial x} \cdot \frac{\partial p}{\partial y} \Rightarrow \rho$$

Poisson 2D:

$$\iint_{\Delta x \Delta y} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) dx dy = \int_{\Delta x \Delta y} \left(\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} \right) dx dy$$

$$\int_{\Delta y} \left[\frac{\partial p}{\partial x} \right] dy + \int_{\Delta x} \left[\frac{\partial p}{\partial y} \right] dx$$

$$= \left[\frac{p_{i+1,j} - p_{i,j}}{\Delta x} - \frac{p_{i,j} - p_{i-1,j}}{\Delta x} \right] \Delta y +$$

$$\left[\frac{p_{i,j+1} - p_{i,j}}{\Delta y} - \frac{p_{i,j} - p_{i,j-1}}{\Delta y} \right] \Delta x$$

$$= \left[\frac{p_{i+1,j} - 2p_{i,j} + p_{i-1,j}}{\Delta x} \right] \Delta y +$$

$$\left[\frac{p_{i,j+1} - 2p_{i,j} + p_{i,j-1}}{\Delta y} \right] \Delta x \cdot \frac{\Delta t}{5}$$

$$\int_{\Delta y} \left[\mu \right]_{i-1/2}^{i+1/2} dy + \int_{\Delta x} \left[v \right]_{i-1/2}^{i+1/2} dx$$

$$= \left(\mu_{i,j} - \mu_{i-1,j} \right) \Delta y \Delta y + \left(v_{i,j} - v_{i,j-1} \right) \Delta x$$

$$p_{i+1,j} + p_{i-1,j} + \frac{p_{i,j+1} + p_{i,j-1}}{\Delta y^2} - \left(\frac{2}{\Delta x^2} + \frac{2}{\Delta y^2} \right) p_{i,j}$$

$$= \left[\frac{u_{i,j}^2 - u_{i-1,j}^2}{\Delta x} + \frac{v_{i,j}^2 - v_{i-1,j}^2}{\Delta y} \right] \frac{\Delta t}{2}$$

point diagonal system

$$\begin{pmatrix} \text{point diagonal system} \end{pmatrix} (p') = (v')$$

Pressure Correction

$$\int_{0 \times y}^{\infty} u = \mu - \frac{\Delta t}{g}$$

$$u^{n+1} = \mu - \frac{\Delta t}{g} \frac{\partial p'}{\partial x}$$

$$\int_{0 \times y}^{\infty} u \frac{\partial u}{\partial y} = \int_{0 \times y}^{\infty} \mu \frac{\partial u}{\partial y} - \frac{\Delta t}{g} \int_{0 \times y}^{\infty} \frac{\partial p'}{\partial x} \frac{\partial u}{\partial y}$$

$$= \mu \frac{\partial u}{\partial y} - \frac{\Delta t}{g} \left(\rho_{i+1} - \rho_i \right) \Delta y$$

$$u^{n+1} = \mu - \frac{\Delta t}{g} \frac{\rho_{i+1} - \rho_i}{\Delta x}$$

$$u^{n+1} =$$

$$u - \frac{\Delta t}{g} \frac{\partial p'}{\partial y} \Big|_{0 \times y} \int \int dx dy \Rightarrow$$

$$u^{n+1} = u - \frac{\Delta t}{g} \frac{\rho_{i,j+1} - \rho_{i,j}}{\Delta y}$$

Always check RHS As per D of mod for description free now

Under given is 10 to 100

DATUM/DATE

Red don't or mostly red from background

Small losses a bit behind the research the growth (in p.m. top)

3) Some get cut

3) Wikipedia OK

PDF

4) Small hypervisors in

1) shows some aspects

2011 (from 2078 (mobi))

4) get cut - 1 with in 2 months

$$Z = \frac{w}{R_0}$$

2) Better or bit up High Speed Camera, Lightness

3) 100

0.00

0.002

Bayesian

2020

8) Explain good better Angel poems.

Apparatus

Explanation 100 Object... Function

Sato

Trialists

0

From long observation

Q - How did you choose initial price for P.O.C. make avoid bad memory